

Crowd
state

Robot
observations

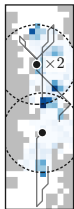
Reconstruction
methods

Output

Evaluation



x_t



y_t, Ω_t

 Senseiver-A	t only
 Senseiver-G (ours)	last 16 s
 DINCAE*	$t \pm 1$ s
 4DVarNet	200 s
 4DVarNet aug. var. (ours)*	200 s
 EnKF*	all past

right: observations used for frame t

* also predicts its uncertainty $\hat{\sigma}_t$



$\hat{x}_t$



$\hat{\sigma}_t$

(example: DINCAE)



accuracy	RMSE
uncertainty	CRPS
	spread/RMSE
speed	ms per frame
on unobserved walkable cells	

ground truth (test days)